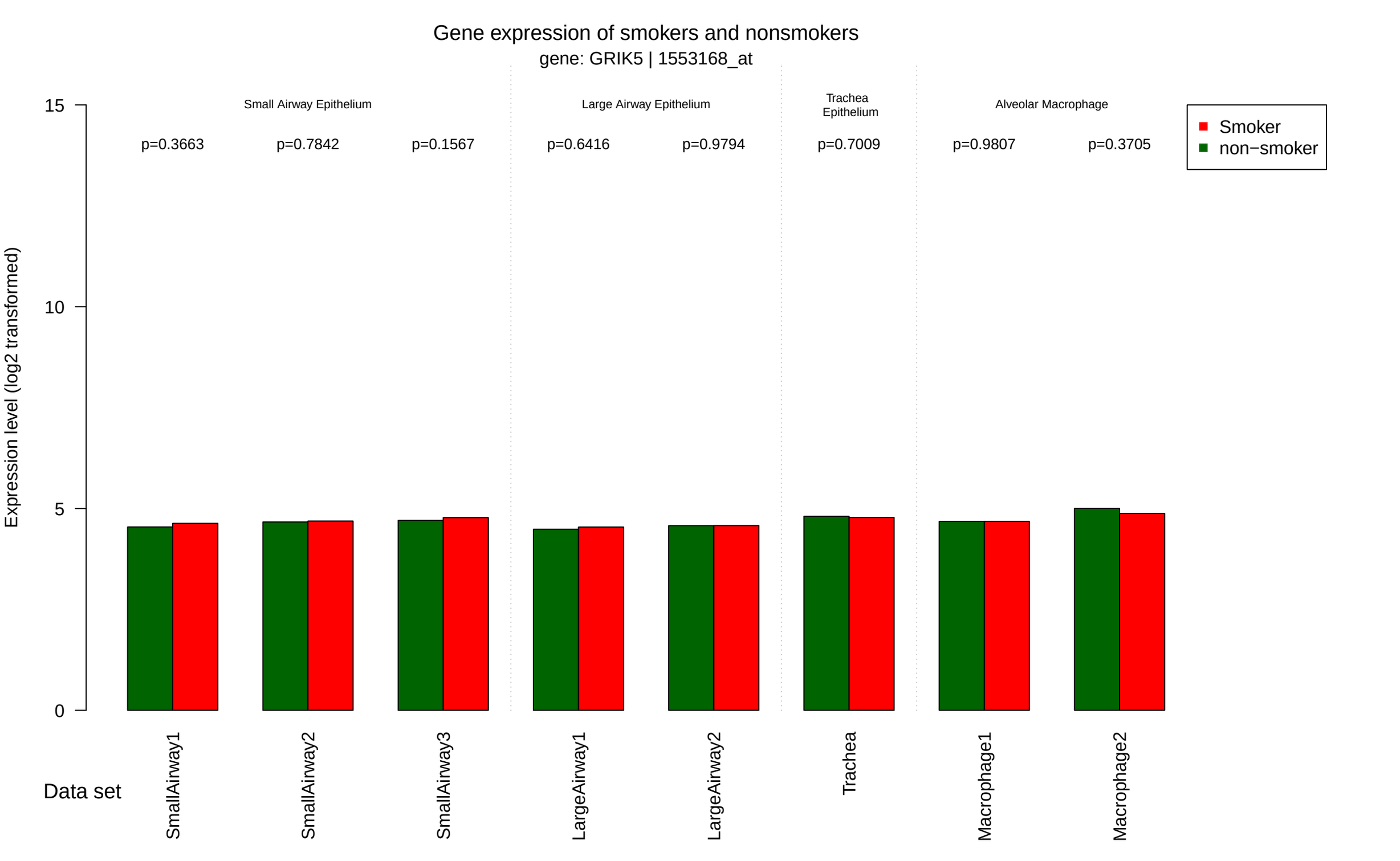
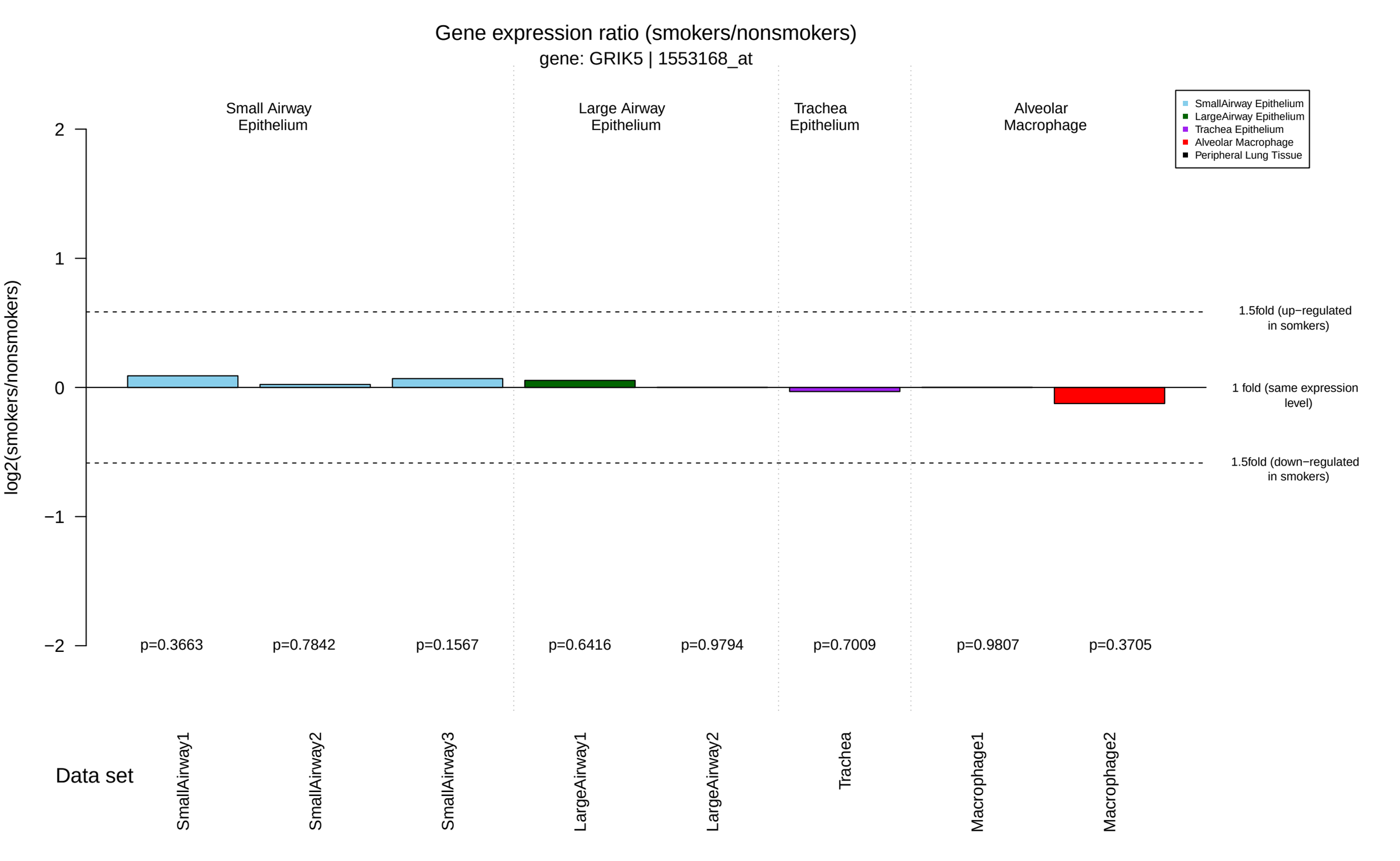
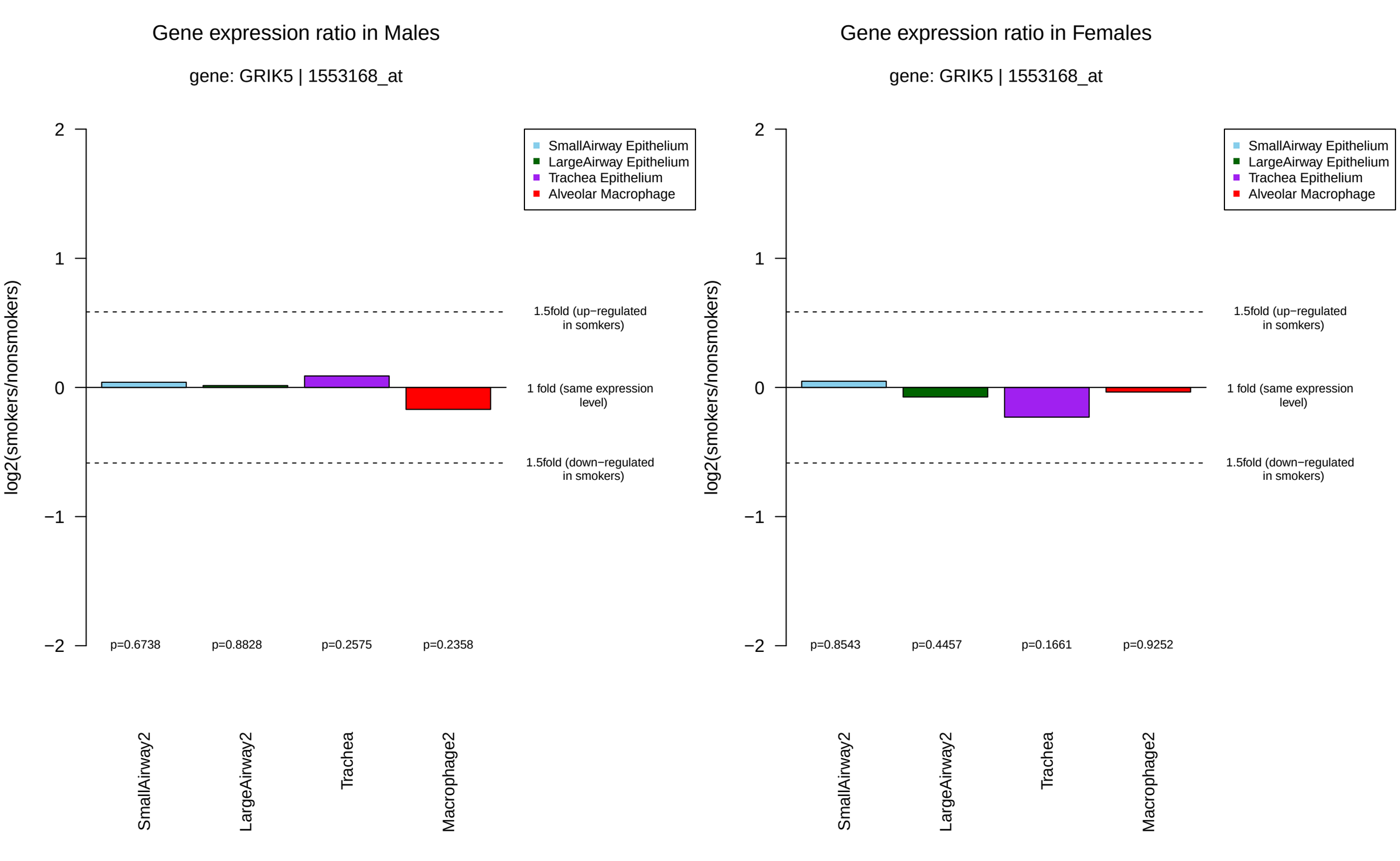
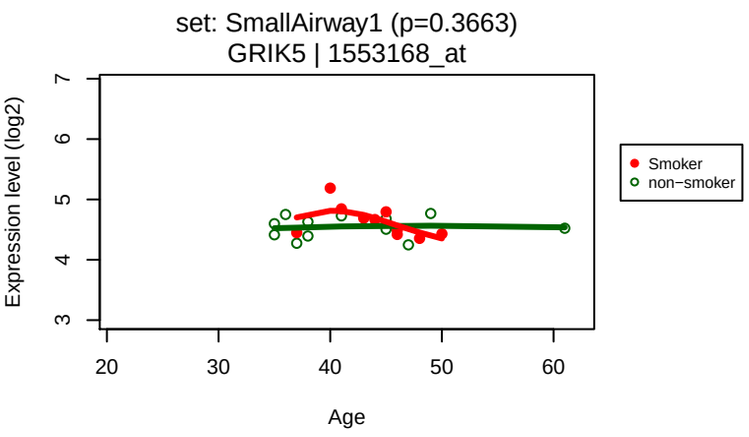
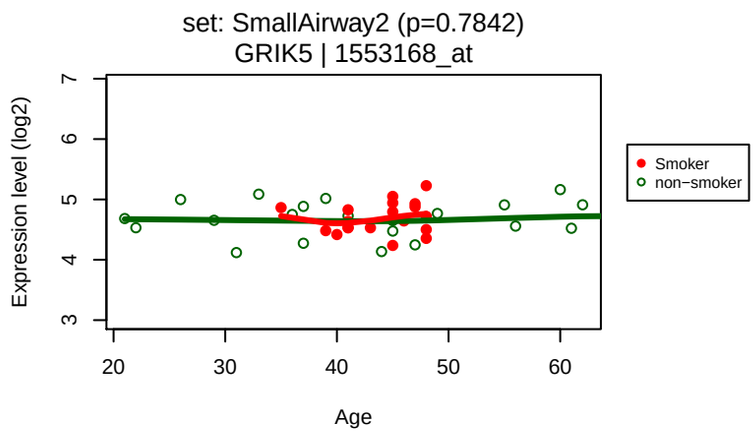
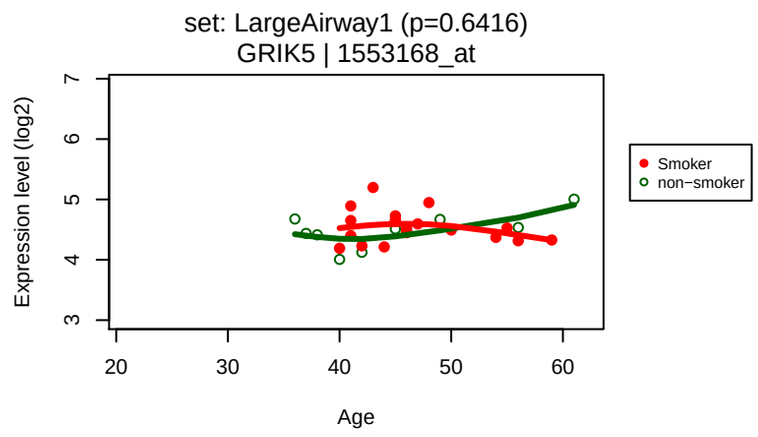

Gene expression of smokers and nonsmokers
gene: GRIK5 | 1553168_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3663
p=0.7842
p=0.1567
p=0.6416
p=0.9794
p=0.7009
p=0.9807
p=0.3705
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

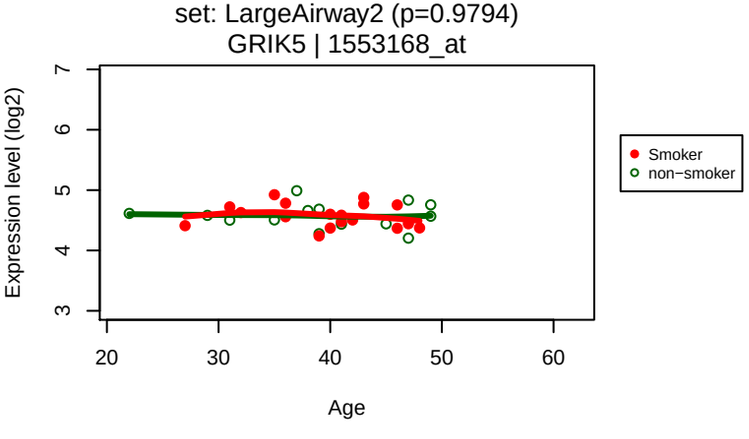

Gene expression ratio (smokers/nonsmokers)
gene: GRIK5 | 1553168_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3663
p=0.7842
p=0.1567
p=0.6416
p=0.9794
p=0.7009
p=0.9807
p=0.3705
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

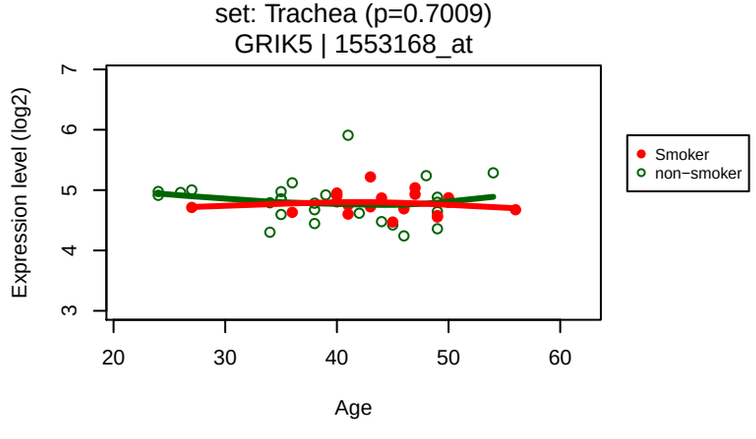

Gene expression ratio in Males
gene: GRIK5 | 1553168_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6738
p=0.8828
p=0.2575
p=0.2358
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIK5 | 1553168_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8543
p=0.4457
p=0.1661
p=0.9252
SmallAirway2
LargeAirway2
Trachea
Macrophage2

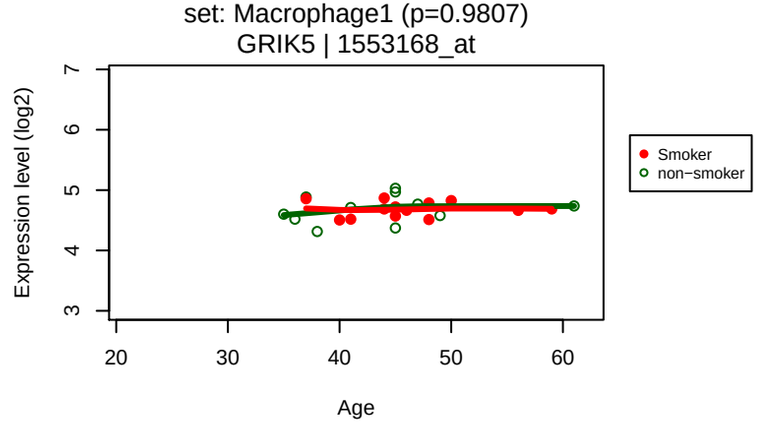

set: SmallAirway1 (p=0.3663)
GRIK5 | 1553168_at
Expression level (log2)
Age
Smoker
non−smoker

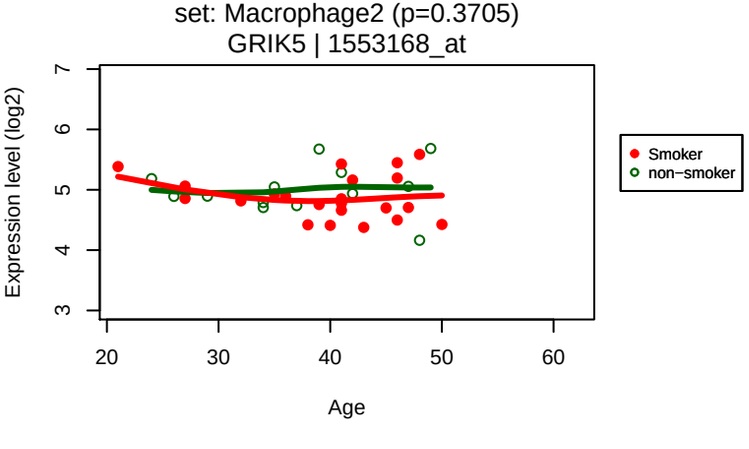

set: SmallAirway2 (p=0.7842)
GRIK5 | 1553168_at
Expression level (log2)
Age
Smoker
non−smoker

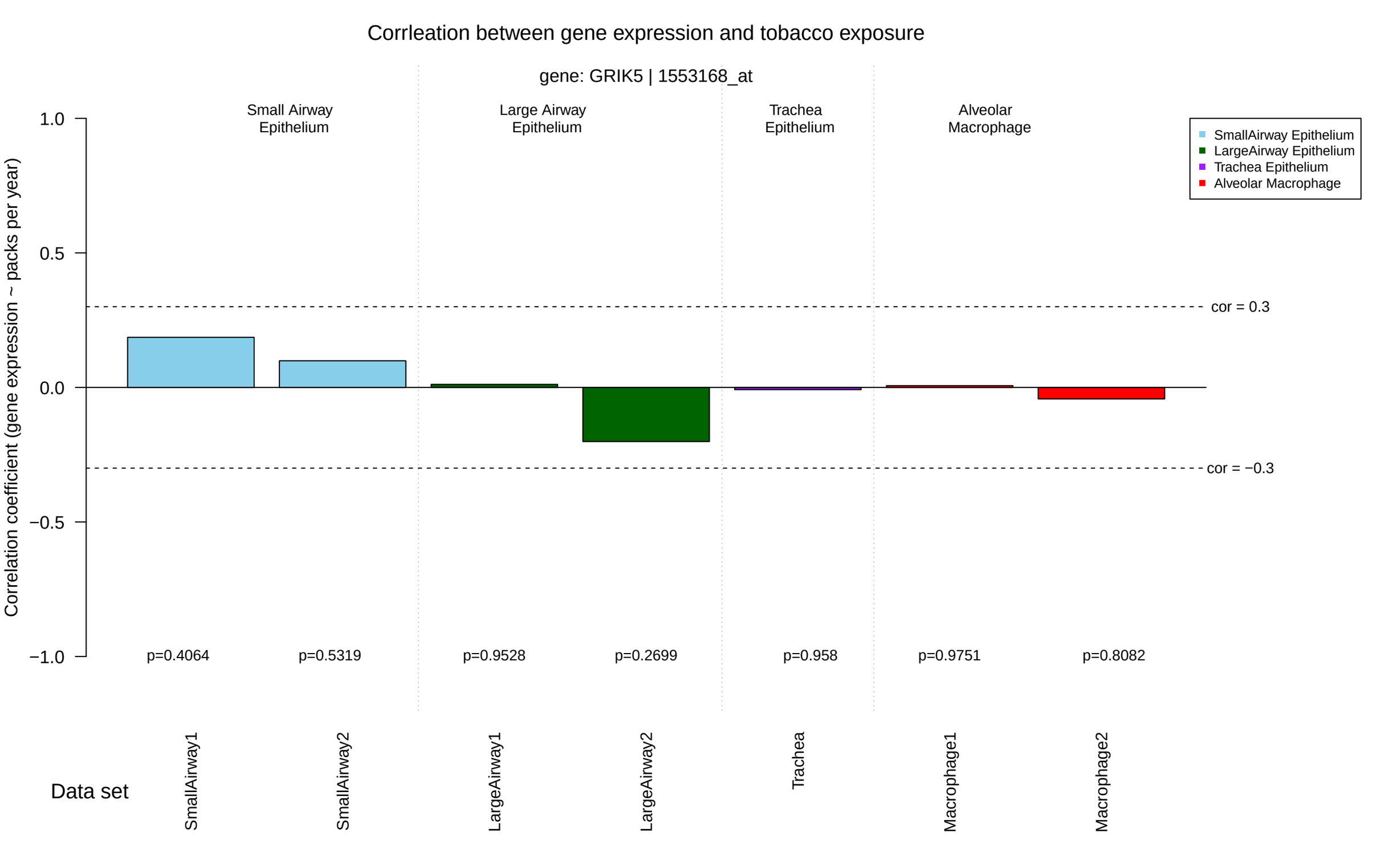

set: LargeAirway1 (p=0.6416)
GRIK5 | 1553168_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9794)
GRIK5 | 1553168_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7009)
GRIK5 | 1553168_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9807)
GRIK5 | 1553168_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3705)
GRIK5 | 1553168_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIK5 | 1553168_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4064
p=0.5319
p=0.9528
p=0.2699
p=0.958
p=0.9751
p=0.8082
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2